

Semi-supervised Identification of Drivers Using only GPS Trajectory Data

Dor Bar
Industrial Engineering
Tel Aviv University
Tel Aviv, Israel
dorbar1@mail.tau.ac.il

Irad Ben Gal
Industrial Engineering
Tel Aviv University
Tel Aviv, Israel
bengal@tauex.tau.ac.il

Abstract—Understanding driving behavior through GPS data offers significant potential for applications such as driver identification and adaptive transportation systems. We introduce CPUT (Contextual Pattern-bagging of Unlabeled Trips), a novel semisupervised framework that clusters drivers based on behavioral patterns extracted from their GPS trajectories. CPUT achieves 94% ACC in clustering and 98% in classification in a case study of a private vehicle, demonstrating its effectiveness in real-world scenarios. Unlike previous works, our approach eliminates the need for labeled data or internal vehicle sensors, making it scalable for large-scale deployment in insurance, fleet management, and mobility behavior analytics.

Index Terms—Driver Identification, GPS Trajectory Data, Semi-Supervised Learning, Active Learning, Unsupervised Clustering, Feature Extraction, Naturalistic Driving Data, Time Series Analysis, Behavioral Pattern Recognition, Telematics

1 INTRODUCTION

The rapid advance of smart transportation technologies is transforming traffic management, autonomous driving, and driver identification. Because GPS traces can be collected not only from vehicle sensors but also from smartphones, aftermarket dongles, and other connected devices, they offer an unprecedented opportunity for advanced driver analytics, allowing the extraction of distinctive driving ‘fingerprints’ directly from raw trajectories (Xun et al. [1]; Halim et al. [2]). Unlike traditional methods that typically rely on internal vehicle sensors or manually annotated datasets, which are often costly, complex, and restricted to controlled settings, GPS-based approaches offer greater accessibility and scalability (Azadani and Boukerche [3]). These digital driving fingerprints can substantially enhance applications such as behavior-based risk assessment (Wang et al. [4]), theft protection through driver authentication (Xun et al. [1]), personalized driver assistance systems, and even autonomous vehicle operations.

Although much of the research in this domain has focused on multiclass identification, these methods often rely on prior knowledge and driver labeling, an assumption that may not be feasible for all real-world applications. Our work seeks to address this limitation by utilizing naturalistic driving data and focusing exclusively on GPS coordinates, eliminating the dependency on additional sensors. traditionally captured

through internal sensors, in effective driver classification. By simplifying data requirements, our research opens new possibilities for scalable and practical automotive analytics.

Our study introduces a robust framework designed to advance the field of driver identification by relying exclusively on GPS data, eliminating the need for additional sensors or pre-labeled datasets. This simplicity enhances the framework’s scalability and applicability across diverse vehicle types and driving conditions, making it highly adaptable for real-world use. To address the challenge of limited labeled data, we incorporate semi-supervised learning techniques, leveraging pattern detection from geospatial information. Our approach extracts meaningful features from GPS data, ensuring a robust representation for downstream tasks. By employing effective clustering strategies, we enhance the accuracy of the identified driving patterns. We structure driver groupings dynamically, refining clusters iteratively to improve identification performance. By combining efficient feature extraction with adaptive clustering, our research delivers a scalable and accurate solution for unsupervised driver identification, offering a practical and generalizable approach for real-world implementation.

Designing an unsupervised pipeline under such constraints introduces several key challenges. Environmental confounders such as traffic conditions, road types, and weather can obscure behavioral signals, making it harder to attribute driving patterns to individuals. The sparse sampling of GPS trajectories - often logged at 0.1 Hz - misses fine-grained maneuvers, reducing the resolution of behavioral cues. Moreover, feature inconsistency poses a problem: variables like speed or jerk may differentiate some drivers well but prove uninformative in others. The data is also imbalanced, as some individuals contribute many more trips than others, biasing clustering outcomes. Lastly, trip variability in route, purpose, and duration further complicates attempts to extract global driver fingerprints. These challenges collectively shape the design decisions in our proposed framework.

When implemented over a single vehicle case study, the CUPT achieved a clustering accuracy of 94 % and an end-to-end AUC of 0.98.

This paper is organized as follows: Section 2 reviews related works on driver identification and the limitations of existing sensor-based and supervised methods. Section 3 details our

proposed methodology, including pre-processing, feature extraction, and the semi-supervised clustering framework. Section 4 presents experimental results along with the sensitivity analysis and discusses the implications of our findings for future smart transportation systems. Finally, Section 5 presents our conclusions and future work.

2 RELATED WORKS

The evolution of driver identification research reveals a paradigm shift from sensor-dependent methodologies to GPS-centric approaches, mirroring the broader adoption of telematics in transportation analytics. Early pioneering work by Eren et al. [5] demonstrated that even basic smartphone sensors could distinguish safety-oriented driving styles through controlled experiments, although their binary classification of "safe" versus "risky" behaviors left finer distinctions unexplored. As GPS technology became ubiquitous, researchers began encoding complete trips rather than isolated maneuvers. Dong et al. [6] made significant strides with a hybrid autoencoder-regularized recurrent network that learned latent style vectors from high-frequency GPS traces, though their unreleased code and private datasets hindered independent validation.

Concurrent developments in vehicle sensor analytics pushed accuracy boundaries through deep embeddings. Hallac et al. [7] achieved 83% pairwise identification accuracy using temporal convolutional networks on multimodal CAN-bus streams, while Wen et al. [8] combined eye-tracking data with vehicle controls to reach near-perfect accuracy in simulator environments. These sensor-heavy approaches, though technically impressive, faced scalability challenges in real-world fleets with heterogeneous instrumentation.

The field gradually pivoted toward sparse GPS analytics as researchers proved motion patterns alone could reveal driver fingerprints. Wang et al. [4] broke new ground with spatiotemporal representation learning that modeled peer and temporal dependencies across trajectory segments. Their work anticipated later sequence modeling approaches but relied on proprietary data from ride-hailing platforms. Lu [9] subsequently demonstrated that regularized recurrent networks could outperform multimodal baselines using only GPS-derived features, though their supervised approach required extensive labeled data.

Parallel advancements in time-series processing enabled more efficient GPS analytics. The MiniROCKET transform [10] replaced learned convolutional filters with deterministic random kernels, accelerating feature extraction by orders of magnitude while maintaining discriminative power. When combined with principal component analysis [11] and density-based clustering [12], this pipeline offered a lightweight alternative to deep sensor models - a critical enabler for large-scale deployment.

Recent years witnessed two competing trends: increasingly complex neural architectures versus interpretable statistical approaches. Azadani et al. [3] and Zheng et al. [13] pushed

accuracy boundaries with siamese temporal convolutional networks, while Banerjee et al. [14] showed simple statistical features could authenticate drivers through careful GPS analysis. Lyu et al. [15] introduced mutual reinforcement between road and driver clustering, and Li et al. [16] achieved state-of-the-art results on private fleet data through hierarchical attention mechanisms. However, these advances came with caveats - proprietary datasets, unreleased codebases, and computational demands that limit real-world applicability.

The present study emerges at this methodological crossroads. By combining MiniROCKET's computational efficiency [10] with context-aware trip normalization and pseudo-labeling, our framework achieves 94% clustering accuracy without requiring sensor fusion or pre-labeled datasets. This bridges the gap between Lu's GPS-only supervision [9] and Banerjee's interpretable features [14], while outperforming both through semi-supervised learning. Crucially, our open implementation using public geocoding APIs [17] and reproducible clustering parameters [18] addresses the transparency issues plaguing recent neural approaches [13], [16].

Three key insights from traffic flow theory [19], [20] underpin our approach: First, that driver behavior manifests through consistent interactions with roadway physics; second, that trip purpose and time-of-day induce measurable routine patterns [21]; third, that queuing dynamics [22] leave identifiable traces in sparse GPS logs. By encoding these principles into destination-time grouping and context-aware scaling, we extract stable behavioral signatures from noisy 0.1Hz trajectories—a capability previously thought to require 10Hz localization [13].

The framework's success highlights an underappreciated truth in telematics: With appropriate feature engineering and semi-supervision, sparse GPS data contains sufficient signal for precise driver identification. This contradicts assumptions underlying both sensor fusion approaches [7], [8] and high-frequency GPS models [2], [13], while aligning with Wang et al.'s [4] findings on spatiotemporal regularity. Our results suggest the field may have overinvested in complex architectures at the expense of scalable, transparent pipelines—a course correction enabled by recent advances in randomized time-series transforms [10] and robust clustering [18] [23].

3 PROPOSED METHOD

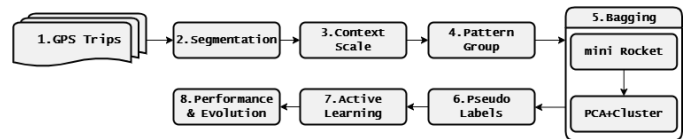


Fig. 1: Architecture diagram.

This section presents our semi-supervised framework for driver identification, built entirely on GPS trajectory data. The pipeline, shown in Figure 1, consists of three main stages: pre-process (1.1 - 1.4), clustering (2.1 - 2.2), deployment and evaluation (3.1 - 3.2).

The process begins with raw GPS trips, which are segmented into short, overlapping windows to capture local driving behavior. These segments are then normalized using a context-aware scaling procedure that accounts for road speed, time-of-day, and trip length. To reduce behavioral noise, we group trips by destination-time patterns, isolating recurring routines that are likely driver-specific.

In the bagging stage, we apply the MiniROCKET transform repeatedly with randomized seeds to extract diverse time-series features. Dimensionality reduction (via PCA) is applied to each feature set, followed by clustering. The resulting cluster assignments are merged to generate robust pseudo-labels that reflect consistent behavioral groupings across multiple feature perspectives.

Finally, we apply a second round of clustering on the aggregated embeddings to refine driver groupings. These serve as soft labels for downstream evaluation, providing a practical and scalable approach to driver identity modeling in the absence of ground truth.

3.1 GPS Trips

The data for this study originate from a major international leasing company that collects detailed driving information as part of its daily operations. Covering the period from 2018 to 2020, the dataset includes naturalistic driving records for more than 8,000 vehicles operating leased vehicles in real-world conditions. These records span a broad range of driving scenarios—diverse routes, road types, and locations, as well as trips undertaken on different days and at various times.

As depicted in table I, the dataset is divided into two primary components. The first component, the GPS dataset, is a time-series collection capturing trip-level details such as vehicle speed, posted speed limits, GPS coordinates, and timestamps. These measurements, recorded at a rate of 0.1 Hz (one sample every 10 seconds), form the core of this study by representing the spatial and temporal characteristics of each trip.

The second component, the Summary dataset, provides an overview of each trip and aids in segmenting the continuous time-series data into individual trips, allowing for a more granular analysis.

A sample of the spatial coverage of these data is presented in Figure 2, which plots GPS coordinates on a map to illustrate the broad geographic extent of the recorded trips. Taken together, these three components supply a comprehensive foundation for investigating driving patterns and behaviors under diverse real-world conditions.

TABLE I: Data Description

Drives	GPS
TIME FROM LAST DRIVE	ROAD SPEED
FUEL USED	VEHICLE STATE
END DRIVE EVENT	TIME
START DRIVE EVENT	LATITUDE
VEHICLE ID	LONGITUDE
DRIVE ID	VEHICLE ID



Fig. 2: Example of a trajectory data of one trip.

3.1.1 Feature Engineering : We use three core features to characterize driving behavior: speed, linear acceleration, and angular acceleration. These are derived directly from the GPS logs and capture dynamic driving patterns while avoiding confounding geographic or temporal biases. Coordinates and timestamps are excluded from the classification to prevent misidentification caused by shared routes or schedules.

Linear acceleration is computed as the centered difference in speed over time, while angular acceleration captures changes in directional movement using a second-order finite difference. To complement these behavioral features, we enrich each trip with contextual geo-information using OpenCage Geocoder [17].

3.2 Segmentation

To improve data quality and behavioral resolution, we segment each GPS trajectory based on stopping points specifically, moments when the vehicle comes to a complete halt. This approach serves multiple purposes: it reduces the influence of external noise sources such as traffic lights and congestion by discarding short, low-informative segments; it supports better normalization in the subsequent scaling step by creating more uniform behavioral units; and it increases the number of usable segments per vehicle, effectively expanding the dataset. This stop-based segmentation allows us to isolate meaningful driving episodes and enhances the model’s ability to capture individual driving patterns.

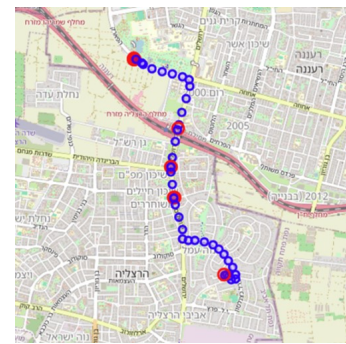


Fig. 3: example of a segmented trip, each red dot represents a timestamp where the speed of the vehicle was zero.

3.3 Context-Aware Scaling

To improve comparability between driving behaviors, we apply a normalization process based on three contextual dimensions: speed limit, time of day, and segment length. These factors are known to influence driving style and help reduce environmental noise caused by traffic, road types, or time-specific congestion.

Each segmented trip is assigned to a contextual bin based on these three attributes. Within each bin, we standardize the key features—speed, linear acceleration, and angular acceleration—by computing their z-scores over all time steps. This localized scaling ensures that comparisons are made between behavior patterns observed under similar external conditions.

To further improve robustness, we detect and remove anomalies in each bin by excluding time series whose feature variances deviate more than three standard deviations from the mean. This filters out outliers caused by GPS noise, sensor drift, or abnormal traffic events.

Importantly, contextual binning is used only during the scaling step. Once normalized, the time series are returned to their original form for downstream processing. This approach balances computational efficiency with behavioral precision, enabling scalable and noise-resilient driver comparisons across diverse environments.

3.4 Pattern Groups

Our method groups trips based on recurring destination-time patterns, under the assumption that repeated travel routines are typically performed by the same driver. Specifically, we group trips by destination and coarse time-of-day categories (morning, afternoon, evening/night), which proved effective for our dataset of private drivers. This structure captures meaningful behavioral patterns while avoiding excessive fragmentation.

To prevent bias from routine commutes, the most frequently visited destination—usually the driver’s home—is excluded from the grouping. This helps highlight distinctive driving behaviors rather than habitual ones.

While our grouping strategy suits naturalistic data from private vehicles, commercial datasets may benefit from finer granularity. Incorporating parameters like exact time slots or work shifts could improve pattern resolution where schedules are more structured and multiple drivers share the same vehicle.

3.5 Bagging

In this section, we introduce the pipeline architecture depicted in Figure 4, which aims to distinguish between pattern groups representing the same driver and those arising from shared or heterogeneous trips. To achieve this, we employ feature bagging, leveraging diverse driving attributes to enhance the robustness of our approach and improve the differentiation between individual and mixed-driver travel patterns.

After segmentation and scaling, we transform each driving segment into a fixed-length feature vector using MiniROCKET [10], a fast and reliable time-series transformation method. It applies a set of random convolutional

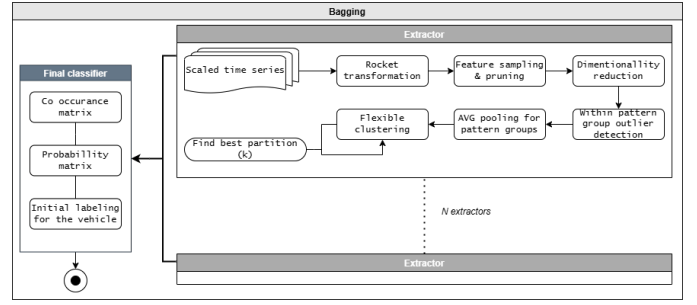


Fig. 4: Feature bagging architecture.

kernels to capture key behavioral patterns such as acceleration trends, steering changes, and speed fluctuations. To improve robustness and increase variability across iterations, we repeat this extraction multiple times, each time sampling 80% of the resulting MiniROCKET features. The final representation is formed by aggregating features across these runs—a process we refer to as feature bagging.

Since the resulting features are high-dimensional, we apply Principal Component Analysis (PCA) to reduce each feature set to a compact representation while preserving the most informative behavioral axes. Because our pipeline includes a contextual scaling step, much of the variability related to external factors (e.g., road type, time of day) is already normalized. As a result, PCA’s variance-based sorting tends to emphasize residual variations that are more closely tied to driver-specific behavior. This not only improves computational efficiency but also enhances the discriminative power and generalization of the downstream clustering. Next, instead of clustering

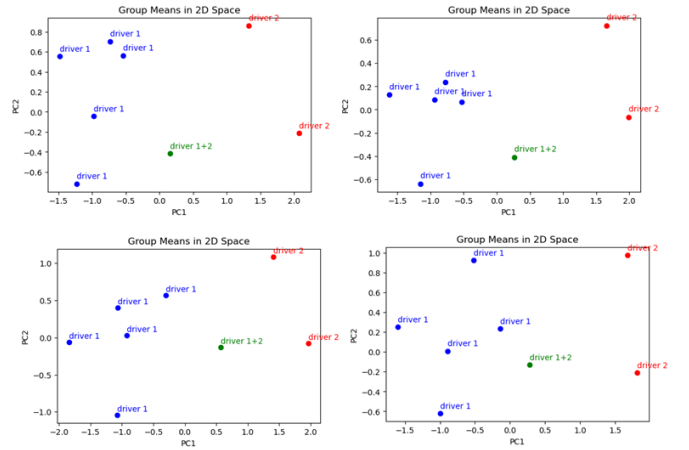


Fig. 5: Group means in a 2D principal component space for four different feature extraction. Each subplot shows the mean positions of three distinct groups—driver 1 (blue), driver 2 (red), and driver 1+2 (heterogeneous group in green)—in the 2D space defined by the first two principal components (PC1 and PC2). These principal components are derived from four different feature extractors (one per subplot).

individual trips, we cluster the pattern groups identified earlier,

ensuring that the analysis focuses on consistent behaviors rather than isolated events. Within each group, we apply DBSCAN to the features of each MiniROCKET-PCA run, both to remove local outliers and to enhance diversity across iterations. Because DBSCAN is sensitive to the geometry of each feature space, this step captures slightly different core samples per iteration - reinforcing the unique characteristics emphasized by each randomized transformation.

Finally, we computed a centroid for each cleaned group and applied a second clustering step, such as spectral clustering, to these centroids. This creates soft and interpretable groupings of driver behavior based on consistent, repeated patterns across the dataset.

3.6 Pseudo Labels

To consolidate clustering results across multiple iterations, we construct a co-occurrence matrix that records how often pattern groups appear together. This captures stable behavioral relationships and helps identify which groups likely belong to the same driver. Applying a threshold to this matrix yields soft driver labels based on consistent co-occurrence patterns.

In practice, we observed that heterogeneous groups, those likely containing trips from multiple drivers, tend to show higher variance across iterations, further validating their mixed composition. These insights support the reliability of our soft labeling process and provide a natural filter for ambiguous cases.

From this point onward, standard supervised models, as proposed in prior works, can be trained on these soft labels to improve prediction quality or support downstream tasks. While our focus is on unsupervised preprocessing and pattern discovery, this framework provides a scalable entry point for supervised driver classification in real-world applications.

3.7 Active Learning

Beginning with the pseudo-labels obtained in the previous stage, we enlarge the labelled pool through an iterative, high-confidence loop. (i) *Prune outliers*. Within every provisional driver cluster we apply DBSCAN to discard anomalous trips. (ii) *Train a probabilistic classifier*. The remaining labeled trips serve as training data for a segment-level model that outputs class probabilities $p_{s,d}$ for each segment s . (iii) *Aggregate to the trip level*. For an unseen trip t the segment probabilities are averaged to obtain a trip-level score $p_{t,d} = \frac{1}{|t|} \sum_{s \in t} p_{s,d}$. (iv) *Select confident trips*. A trip is accepted only if its highest probability lies outside a neutral band, e.g. $p_{t,d} \notin [0.5 - \delta, 0.5 + \delta]$ with $\delta = 0.15$; this ensures that the model “trusts” the assignment. All segments of these high-confidence trips are then added to the labelled set. Steps (ii)–(iv) may be repeated until no additional trips qualify, this step can be taken to strengthen the model and improve it for online learning. the primary focus in this step is to mimic an online-learning scenario.

3.8 Performance & Evaluation

Assessing a driver-identification pipeline in the presence of only fragmentary ground truth is non-trivial, and we therefore combine three complementary evaluation strategies.

When *complete* ground-truth driver IDs are available, performance is usually summarised with three popular, label-aware scores: plain *accuracy* (fraction of correctly assigned segments); *Normalised Mutual Information* (NMI), which measures the mutual dependence between predicted and true partitions while remaining invariant to permutation of cluster labels; and its chance-corrected counterpart *Adjusted Mutual Information* (AMI). Accuracy is intuitive but ignores the underlying cluster structure, whereas NMI and especially AMI provide a more robust indication of how well the algorithm recovers the latent grouping beyond random agreement. Because our dataset contains only partial labels, we cannot rely solely on these supervised metrics and therefore adopt the additional strategies outlined below.

a) *Distribution stability*.: If the pseudo-labels learnt by the active-learning loop are reliable, the *frequency of trips per driver* should remain essentially unchanged after the newly accepted trips are folded into the training set. Monitoring this before-versus-after histogram provides an immediate, label-free sanity check: any dramatic shift would indicate that the model is absorbing systematic errors rather than genuine information.

b) *Segment-level validation*.: In a second test we treat the pseudo-labels obtained during the bagging phase as provisional ground truth, stratify the data by driver, and apply cross-validation with bootstrap resampling. The resulting distribution of segment-level accuracies indicates to what extent the model can consistently re-identify micro-trajectories that it has *not* seen during training, even though the only labels available are self-generated.

c) *Unsupervised accuracy proxy*.: Finally, we rely on an entirely label-agnostic statistic that proved highly correlated with the supervised scores in Table IV. For every vehicle we compute the ratio of mean intra-cluster to mean inter-cluster Dynamic Time-Warping (DTW) distance,

$$\text{Ratio} = \frac{\text{intra-DTW}}{\text{inter-DTW}}, \quad \widehat{\text{Acc}} = \frac{1}{1 + |\text{Ratio} - \text{Ratio}_{\text{ref}}|}, \quad (1)$$

where $\text{Ratio}_{\text{ref}}$ comes from a single manually labelled reference vehicle. Smaller ratios imply tighter, better-separated clusters and thus inflate the estimated accuracy; empirically the estimate trails the observed supervised accuracy by only a few percentage points, providing a practical upper bound in scenarios where labelled data are scarce or absent.

4 EXPERIMENTS AND RESULTS

This section quantifies the performance and robustness of the proposed *GPS-only*, semi-supervised pipeline under four complementary scenarios. We first use the single vehicle for which ground truth is available - 906 trips, segmented into 5 107 trajectory segments - to measure, with 1805 belongs to one driver and the rest to the other one. Absolute accuracy,

examine the benefit of context-aware scaling, and compare against six re implemented baselines. We then test whether pattern groups learned from one calendar year generalize to a second year in a streaming setup driven by a segment-wise autoencoder + XGBoost stack. To probe external validity we apply the unsupervised part of the pipeline to fourteen unlabeled vehicles, infer 32 latent drivers and five behavioural clusters, and evaluate how well a classifier trained on those clusters transfers back to the fully labeled car. Finally, a sensitivity analysis visualizes how ROC-AUC, respond to alternative dimensionality-reduction methods (PCA, t-SNE, UMAP), different component counts and progressive feature-pruning thresholds. Together these tests cover baseline comparison, online applicability, multi-vehicle generalisation and hyper-parameter robustness while operating in the data-sparse, label-limited setting that motivates this work.

4.1 Baselines

We reproduced six baselines - *LOADS*, *ArrNet+PB+NCE*, *Mini-Rocket*, *RJC*, and *TGBDT* - to benchmark our pipeline (Table II). Every algorithm was reconstructed from the literature and has undergone minimal modifications to adapt to our dataset.

Because several baselines rely on dense CAN-bus or detailed road graphs, we relaxed turn-detection thresholds, approximated jerk profiles and omitted map-matching, which explains the modest scores of *LOADS*, *Mini-Rocket* and *RJC*. *ArrNet+PB+NCE* was trained self-supervised with a noise-contrastive loss, after which its embeddings were aggregated into the same pattern groups used by our method. Whenever a baseline produced only a feature embedding, we combined it with four standard clustering algorithms (*k*-means, agglomerative, DBSCAN, spectral) and kept the best result. All hyper-parameters - window length, filter count, distance cut-offs, linkage criteria - were tuned via grid search on a validation split and fixed before testing.

Under these constraints DTW+PB, a Deterministic precursor of our own pipeline, emerges as a strong and simple baseline with a mean accuracy of 0.87. The full system proposed in this study lifts performance to 0.94 with similar or smaller variance, demonstrating the added value of confidence-filtered active learning.

a) *Internal DTW heuristic.*: For completeness we also evaluated a deterministic DTW clustering routine that operates directly on the trip segments and pattern groups generated in Steps 2–4. Although it attains 0.91 accuracy, the method is hard-clustered and cannot resolve heterogeneous driving styles.

Implementation note. The original authors of *LOADS*, *RJC*, and *ArrNet* have not released either code or full datasets. Our versions were reproduced from the algorithmic descriptions only; they replicate the published preprocessing, feature and loss functions, but—not having access to the authors’ tuning logs—may not match the exact hyper-parameter settings that produced the original numbers. The baseline scores reported

TABLE II: Unsupervised baselines on the labelled-car use-case. Numbers in parentheses are standard deviations over repeated runs.

Method	Accuracy	NMI	AMI
RJC (det.) - based [14]	0.644 (0)	0.00035 (0)	0 (0)
DTW+PB (det.)	0.87 (0)	0.756 (0)	0.67 (0)
LOADS - based [16]	0.56 (0.026)	0.01 (0.001)	0 (0.001)
ArrNet - based [6]	0.60 (0.09)	0.445 (0.084)	0.091 (0.013)
Mini-rocket [10]	0.62 (0.10)	0.004 (0.002)	0.002 (0.001)
TGBDT [30]	0.66 (0.092)	0.0056 (0.0003)	0.007 (0.001)
CPUT	0.94 (0.01)	0.98 (0.07)	0.90 (0.097)

det. - deterministic (single-run result).

here should therefore be regarded as indicative rather than a verbatim reproduction of the published results.

4.2 Active Learning

We tested the framework in a streaming setting on a single long-haul vehicle whose log spans two full years. Pattern groups and pseudo-labels were learned from *Year 1* only; no label-refinement loop was executed thereafter. The experiment asks whether those one-year pattern groups already encode a stable driver signature that generalises to all trips in *Year 2*, including trips that had originally been mixed across groups.

A segment-level auto-encoder, architecturally equivalent to *ArrNet*, produced 128-dimensional embeddings. Each embedding, one per trajectory segment, served as a training sample for an XGBoost classifier. At inference time the classifier output class probabilities $p_{s,d}$ for every unseen segment s . Trip-level scores were then obtained by arithmetic mean, $p_{t,d} = \frac{1}{|t|} \sum_{s \in t} p_{s,d}$.

Figure 6 displays the resulting confusion matrix and ROC curve for *Year 2*. The model attains an accuracy of 0.94 and a ROC-AUC of 0.98; misclassification affects only 2.8 % of driver-1 trips and 3.5 % of driver-2 trips. Hence a single year of pattern groups - combined with context scaling, PCA and a segment-wise auto encoder followed by XGB stack, was sufficient to distinguish subsequent trips that

the original hard clustering could not separate. This outcome confirms that the learned pattern groups capture a durable driver signature usable by downstream classifiers without further manual intervention.

4.3 Utilizing Unlabeled Vehicles

In a final experiment, we tested our framework’s scalability in a scenario with minimal labeled data. Fourteen vehicles, with no prior driver labels, underwent our proposed clustering procedure to generate inferred labels solely from the unsupervised methods in Section 3. We discovered 32 distinct drivers

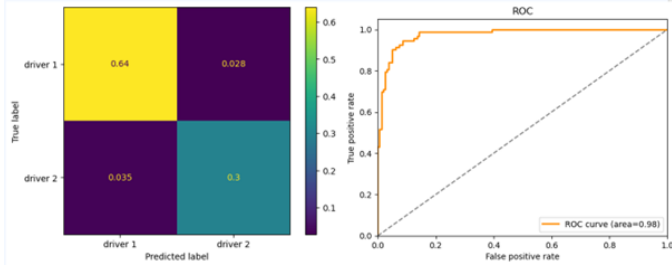


Fig. 6: Results of the active learning over the presented use-case.

among these vehicles and further aggregated them into five behavioral clusters. We then matched these five clusters to our fully labeled drivers based on similarity in driving patterns and trained a classifier on the inferred labels.

a) *Evaluation*:: We tested how well this classifier—trained on purely unlabeled clusters—generalized to our labeled drivers. Table III summarizes four classification models:

Key Observations:

- Random Forest (RF) and XGBoost: Top generalization performance. XGBoost outperforms RF on the test set (AUC = 0.86, Accuracy = 0.77).
- Logistic Regression (LR): Moderate generalization, at Accuracy = 0.72%.
- KNN: Performs the worst (Accuracy $\approx$ 0.53%), suggesting difficulty handling the inherent variability in driver behavior across labeled and unlabeled clusters.
- Tree-based models (XGBoost, RF) best leverage unsupervised clustering in a semi-supervised setting. XGBoost’s superior test performance underscores its robustness to heterogeneity in driver identities.

b) *Unsupervised $\rightarrow$ supervised link*:: Table IV juxtaposes the *silhouette-based accuracy estimates* with the *observed supervised accuracies* for all 15 vehicles. Their near-linear match confirms that the tighter and more separated a vehicle’s clusters are, the easier it is to classify once pseudo-labels are present. The derivation of the DTW ratio that underlies these estimates is given in Sec. 3.H.

4.3.1 *Driver-Identification Experiment*: The same 15 vehicles are processed through a *MiniRocket* feature extractor, dimensionality reduction, and an XGBoost classifier. The evaluation—five-fold cross-validation repeated with B bootstrap resamples per fold—yields the mean segment-level accuracies and 95% confidence intervals listed in Table IV. All accuracies are computed with respect to the pseudo-labels produced by our unsupervised pipeline, not ground-truth driver IDs.

The labelled reference vehicle (ID 0) provides a benchmark of roughly 0.80 accuracy. Several vehicles, notably IDs 6 and 13, exceed 0.95 accuracy with narrow intervals, demonstrating that their speed and acceleration profiles are highly distinctive. In contrast, vehicle 4 reaches only 0.72 accuracy and exhibits a wider interval, suggesting overlapping driving

styles or limited data. Overall, the experiment shows that GPS-derived dynamics, once pseudo-labelled, support reliable driver identification, and that the quality of the initial clustering is a strong predictor of the downstream classifier’s success.

4.4 sensitivity analysis

a) *Sensitivity of ROC-AUC to dimensionality-reduction method and component count*: Figure 7 compares driver-classification ROC-AUC as we vary the embedding technique (PCA, t-SNE, UMAP) and the number of retained components (2 – 15). PCA clearly dominates, peaking at **0.90–0.92** with only 5–8 components and degrading slightly beyond that point, while t-SNE plateaus around **0.73** and UMAP around **0.48–0.50**. These results confirm that (i) most discriminatory power is concentrated in the first few principal axes of the *MiniRocket* feature space, and (ii) nonlinear manifolds bring no added value for this task.

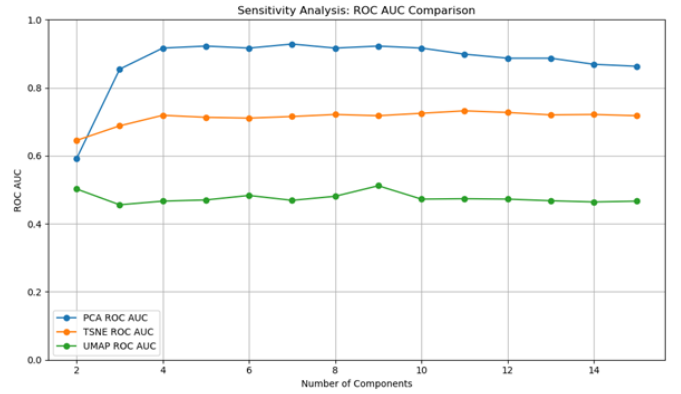


Fig. 7: Sensitivity analysis of dimensionality-reduction methods.

b) *Effect of feature-space filtering on cumulative ROC-AUC*: Figure 8 tracks the cumulative ROC-AUC over 30 bootstrap iterations with 50 extractors for 5-component (left) and 10-component (right) embeddings under four filtering regimes: none, high-variance removal (*std*), correlation pruning (*corr*), and both. Correlation pruning alone already yields rapid convergence to $\approx$ 0.92, while variance-based filtering helps only marginally; combining both attains the same ceiling but with smaller run-to-run variance. Importantly, the “none” curve saturates at $\approx$ 0.8–0.92, demonstrating that eliminating redundant kernels is the key driver of the accuracy gain. we can see that as we add extractor the process becomes more stable.

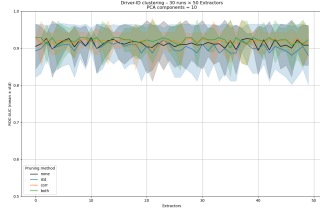
c) *Scaling effect*: To evaluate the impact of normalization on driver identification, we repeated the same procedure on both normalized and non-normalized datasets. Normalization aims to mitigate environmental biases, such as road conditions, traffic patterns, and time-of-day effects, which can substantially influence driving behavior. By standardizing speed, acceleration, and angular acceleration relative to similar road segments, we aimed to ensure that any observed differences stem from the driver’s style rather than external factors.

TABLE III: Classifier Performance on Labeled Data. The models were trained on two clusters of drivers extracted using our method, with each cluster closely matching the corresponding driver in the labeled data.

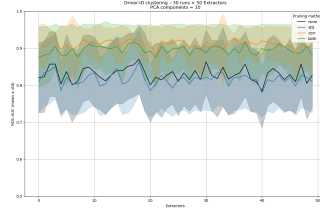
Classifier	AUC	Train				Test			
		Recall	Accuracy	Precision	AUC	Recall	Accuracy	Precision	
RF	1.00	1.00	1.00	1.00	0.81	0.78	0.70	0.56	
XGB	0.99	0.93	0.96	0.97	0.86	0.80	0.77	0.65	
LR	0.93	0.91	0.92	0.91	0.75	0.75	0.72	0.58	
KNN	1.00	1.00	1.00	1.00	0.59	0.75	0.52	0.41	

TABLE IV: Unsupervised clustering quality ($B=1000$) versus supervised classification accuracy ($B=500$).

Veh. ID	#Drv	Clustering metrics				Classification metrics		
		Silh.	Est. Acc.	CI _{low}	CI _{high}	Mean Acc.	CI _{low}	CI _{high}
0	2	0.024	0.940	0.870	0.970	0.802	0.787	0.817
1	2	0.0018	0.920	0.616	0.972	0.809	0.786	0.832
2	2	0.0033	0.896	0.570	0.951	0.894	0.883	0.905
3	2	0.0178	0.927	0.524	0.952	0.835	0.816	0.852
4	3	0.0089	0.961	0.660	0.954	0.717	0.701	0.732
5	2	0.0106	0.951	0.835	0.997	0.797	0.783	0.812
6	2	0.0067	0.889	0.792	0.865	0.984	0.981	0.987
7	2	0.0206	0.911	0.565	0.948	0.820	0.804	0.835
8	2	0.0301	0.894	0.767	0.987	0.781	0.755	0.804
9	2	0.0042	0.922	0.559	0.957	0.782	0.768	0.795
10	2	0.0308	0.948	0.844	0.995	0.818	0.806	0.831
11	2	0.1051	0.849	0.716	0.984	0.915	0.903	0.926
12	2	0.0352	0.946	0.836	0.971	0.801	0.780	0.822
13	2	0.0454	0.906	0.883	0.972	0.949	0.941	0.955
14	3	-0.0139	0.795	0.582	0.899	0.800	0.786	0.811
Avg	2.13	0.0220	0.910	0.708	0.958	0.834	0.819	0.848



(a) 5 components



(b) 10 components

Fig. 8: Sensitivity analysis of feature-pruning strategies—cumulative ROC-AUC across iterations for four settings: no filtering (blue), standard-deviation filtering only (orange), correlation pruning only (green), and the combined variance + correlation filter (red).

Comparing these two datasets allows us to test whether normalization enhances the model’s ability to differentiate between drivers or whether it potentially removes critical behavioral signals. If normalization is beneficial, improved clustering coherence and classification accuracy are expected. Conversely, if raw data performs better, it may indicate that inherent driving patterns suffice for driver differentiation and that normalization introduces unnecessary abstraction.

Table V shows that, without scaling, Spectral Clustering, k -means, and GMM converge to a weighted accuracy of ≈ 0.504 , failing to capture the heterogeneous nature of one pattern group and merging most drivers into a single cluster. Agglomerative Clustering demonstrates modest improvement (accuracy ≈ 0.608), while K-medoids performs poorly at

TABLE V: Clustering Results leaving the contextual scaling and the pattern groups separately

Metric	Full Method		-Con. Scaling		-Pattern groups	
	Acc	NMI	Acc	NMI	Acc	NMI
K-means	0.867	0.829	0.504	0	0.53	0.24
GMM	0.867	0.829	0.504	0	0.58	0.233
K-medoids	0.604	0	0.310	0	0.52	0.01
Spectral	0.94	0.98	0.504	0	0.62	0.37
Agglomerative	0.94	0.98	0.608	0.274	0.59	0.19

accuracy ≈ 0.3 .

When scaling is applied, both Spectral and Agglomerative Clustering reach good performance (accuracy = 0.94), and k -means and GMM also show significant improvement (accuracy ≈ 0.867) but still misassign the heterogeneous case. K-medoids improves to accuracy = 0.604, highlighting its sensitivity to feature scaling. Without scaling, most algorithms fail to distinguish more than one cluster, indicating that scaling is crucial in revealing meaningful driving-pattern distinctions.

d) *Pattern Detection effect*: We next evaluated driver identification performance when omitting the intermediate pattern-detection step (i.e., clustering trips directly). This setup reveals the intrinsic difficulty of driver identification without the benefit of grouping.

5 CONCLUSION & OUTLOOK

This paper introduced a purely GPS-based, semi-supervised framework that identifies drivers with **0.94** accuracy and an AUC of **0.98**, despite the sparse sampling rate of 0.1 Hz

and the absence of auxiliary sensors. The approach combines pattern-group, segmentation, context-scaling, principal-component reduction and MiniRocket feature bagging; this combination was sufficient to emphasise each driver's global behavioural signature and to isolate it from road-environment artefacts.

Although the entire evaluation used data from a single vehicle, the resulting metrics already match or exceed state-of-the-art figures reported for multi-sensor studies. A lightweight context-scaling heuristic removed enough environmental noise for driver-specific signals to emerge inside the pattern groups, confirming that elaborate filtering is not always necessary at sub-hertz sampling rates.

Future work will focus on three directions. First, the environmental filter will be sharpened by injecting external layers—such as weather conditions or real-time traffic—so that subtle driver cues are preserved while context effects are further suppressed. Second, the framework will be validated on heterogeneous commercial fleets to establish robustness and scalability under operational constraints. Third, we plan to quantify the extent to which each pattern group represents a distinct driver persona and to refine the grouping logic by learning dedicated embeddings for those groups. Together, these steps aim to extend the industrial relevance of GPS-only driver identification while maintaining the method's minimal sensing requirements.

REFERENCES

- [1] Y. Xun, J. Liu, N. Kato, Y. Fang, and Y. Zhang, "Automobile driver fingerprinting: A new machine learning based authentication scheme," *IEEE Trans. Ind. Informat.*, vol. 16, pp. 1417–1426, 2020, doi: 10.1109/TII.2019.2946626.
- [2] Z. Halim, R. Kalsoom, and A. R. Baig, "Profiling drivers based on driver dependent vehicle driving features," *Appl. Intell.*, vol. 44, pp. 645–664, 2016.
- [3] M. N. Azadani and A. Boukerche, "Siamese temporal convolutional networks for driver identification using driver steering behavior analysis," *IEEE Trans. Intell. Transp. Syst.*, pp. 1–12, 2022, doi: 10.1109/TITS.2022.3151264.
- [4] P. Wang, X. Li, Y. Zheng, C. Aggarwal, and Y. Fu, "Spatiotemporal Representation Learning for Driving Behavior Analysis: A Joint Perspective of Peer and Temporal Dependencies," *IEEE Trans. Knowl. Data Eng.*, vol. PP, pp. 1–1, 2019, doi: 10.1109/tkde.2019.2935203.
- [5] H. Eren, S. Makinist, E. Akin, and A. Yilmaz, "Estimating driving behavior by a smartphone," in *Proc. IEEE Intell. Vehicles Symp.*, 2012, pp. 234–239, doi: 10.1109/IVS.2012.6232137.
- [6] W. Dong, T. Yuan, K. Yang, C. Li, and S. Zhang, "Autoencoder regularized network for driving style representation learning," *arXiv preprint arXiv:1701.01272*, 2016. [Online]. Available: <https://arxiv.org/abs/1701.01272>
- [7] D. Hallac, Y. Park, J. Leskovec, and S. Saab, "Driver2vec: Machine learning embeddings for automotive applications," *IEEE Trans. Intell. Transp. Syst.*, 2021.
- [8] H. Wen, X. Ma, R. Tao, Y. Yang, C. Zhu, and Q. Yao, "Human-Factors-in-Driving-Loop: Driver identification and verification via a deep learning approach using psychological behavioral data," *IEEE Trans. Hum.-Mach. Syst.*, 2021.
- [9] L. Lu, "Learning driving style embedding from GPS-derived moving patterns for driver identification," *arXiv preprint arXiv:2401.06986*, 2024. [Online]. Available: <https://arxiv.org/abs/2401.06986>
- [10] A. Dempster, D. F. Schmidt, and G. I. Webb, "MINIROCKET: A very fast (almost) deterministic transform for time series classification," in *Proc. 27th ACM SIGKDD Conf. Knowledge Discovery & Data Mining*, 2021, pp. 248–257, doi: 10.1145/3447548.3467231.
- [11] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed. New York, NY, USA: Springer, 2002.
- [12] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proc. 2nd Int. Conf. Knowl. Discov. Data Mining*, 1996, pp. 226–231.
- [13] Y. Zheng, X. Huang, Z. Li, and Y. Lv, "Siamese Multiple Attention Temporal Convolutional Network," *arXiv preprint arXiv:2408.09230*, Aug. 2024. [Online]. Available: <https://arxiv.org/abs/2408.09230> [Accessed: Jun. 7, 2025].
- [14] S. Banerjee, A. Das, and S. Roy, "Driver behavior profiling using machine learning," in *Advances in Communication, Devices and Networking*. Singapore: Springer, 2020, pp. 189–198, doi: 10.1007/978-981-15-8685-9_16.
- [15] Lyu, S., Wang, D., Yang, X. *et al.* Driver profiling using trajectories on arbitrary roads by clustering roads and drivers successively. *Memetic Comp.* 16, 255–267 (2024). <https://doi.org/10.1007/s12293-024-00416-4>
- [16] Li, M., Zhang, Y., Zhao, Y. *et al.* Trajectory Representation Learning with Multilevel Attention for Driver Identification. *Expert Syst. Appl.* 262, 125580 (2025). <https://doi.org/10.1016/j.eswa.2024.125580>
- [17] OpenCage Geocoder, "OpenCage Geocoder API." [Online]. Available: <https://opencagedata.com> [Accessed: Jan. 19, 2024].
- [18] A. Starczewski, P. Goetzen, and M. J. Er, "A new method for automatic determining of the DBSCAN parameters," *J. Artif. Intell. Soft Comput. Res.*, vol. 10, no. 3, pp. 209–221, 2020.
- [19] D. Helbing, "Traffic and related self-driven many-particle systems," *Rev. Mod. Phys.*, vol. 73, no. 4, pp. 1067–1141, 2001.
- [20] D. Ni, *Traffic Flow Theory: Characteristics, Experimental Methods, and Numerical Techniques*. Oxford, U.K.: Butterworth-Heinemann, 2015.
- [21] H. S. Mahmassani and R. Herman, "Dynamic user equilibrium departure time and route choice on idealized traffic arterials," *Transp. Sci.*, vol. 18, no. 4, pp. 362–384, 1984.
- [22] M. Treiber and A. Kesting, *Traffic Flow Dynamics: Data, Models and Simulation*. Heidelberg, Germany: Springer, 2013.
- [23] A. Y. Ng, M. I. Jordan, and Y. Weiss, "On spectral clustering: Analysis and an algorithm," in *Proc. 15th Int. Conf. Neural Inf. Process. Syst. (NIPS'01)*, Vancouver, BC, Canada, 2001, pp. 849–856.
- [24] D. Hallac, J. Leskovec, and S. Boyd, "Driving style embedding with deep learning," *arXiv preprint arXiv:1609.07415*, 2016. [Online]. Available: <https://arxiv.org/abs/1609.07415>
- [25] Y. Wang, Y. Zheng, and Y. Xue, "Travel time estimation of a path using sparse trajectories," in *Proc. 20th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2014, pp. 25–34, doi: 10.1145/2623330.2623656.
- [26] J. Gao, J. Wu, and J. Sun, "Learning traffic as images: A deep convolutional neural network for large-scale transportation network speed prediction," *Sensors*, vol. 18, no. 9, p. 2970, 2018, doi: 10.3390/s18092970.
- [27] G. F. Newell, *Applications of Queueing Theory*, 2nd ed. London, U.K.: Chapman and Hall, 1982.
- [28] X. Wang, W. Cottrell, and S. Mu, "Using k-means clustering to identify time-of-day break points for traffic signal timing plans," in *Proc. 2005 IEEE Intell. Transp. Syst.*, Vienna, Austria, 2005, pp. 586–591, doi: 10.1109/ITSC.2005.1520102.
- [29] K. Fitzpatrick, P. Carlson, M. Brewer, and M. Wooldridge, "Design factors that affect driver speed on suburban arterials," *Transp. Res. Rec.*, no. 1838, pp. 77–85, 2003.
- [30] Kaggle, "Driver Telematics Analysis," 2015. [Online]. Available: <https://www.kaggle.com/c/axa-driver-telematics-analysis/data> [Accessed: Jan. 14, 2015].
- [31] C. C. Aggarwal, *Outlier Analysis*, 2nd ed. Cham, Switzerland: Springer, 2016.
- [32] T. G. Dietterich, "Ensemble methods in machine learning," in *Multiple Classifier Systems*. Berlin, Germany: Springer, 2000, pp. 1–15.
- [33] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Burlington, MA, USA: Morgan Kaufmann, 2011.
- [34] B. Settles, "Active learning literature survey," Univ. Wisconsin–Madison, Dept. Comput. Sci., Tech. Rep. TR1648, 2009. [Online]. Available: <http://digital.library.wisc.edu/1793/60660>
- [35] D. Cacciarelli and M. Kulahci, "Active learning for data streams: a survey," *Mach. Learn.*, vol. 113, pp. 185–239, 2024, doi: 10.1007/s10994-023-06454-2.
- [36] U. Fugiglando, E. Massaro, P. Santi, S. Milardo, K. Abida, R. Stahlmann, K. Netter, and C. Ratti, "Driving Behavior Analysis through CAN Bus Data in an Uncontrolled Environment," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 12, pp. 487–496, 2017.

- [37] D. Ovadia, A. Segal, and N. Rabin, “Classification of hand and wrist movements via surface electromyogram using the random convolutional kernels transform,” *Sci. Rep.*, vol. 14, Art. no. 4134, 2024, doi: 10.1038/s41598-024-54677-7.

6 APPENDIX

DTW cluster technique: For each group i and each trip j :

$$\text{within_AVG}_{i,j} = \frac{\sum_{k=1}^{n[i]} \text{DTW}(\text{drive}_{i,j}, \text{drive}_{i,k})}{n[i]},$$

$$\text{within_AVG}_i = \frac{\sum_{k=1}^{n[i]} \text{within_AVG}_{i,k}}{n[i]},$$

$$\text{STD}_i = \frac{\sum_{k=1}^{n[i]} \left(\text{within_AVG}_{i,k} - \text{within_AVG}_i \right)^2}{n[i] - 1}.$$

Trips deviating by more than $\pm 3 \text{STD}_i$ from the group mean were considered outliers and removed. Next, the average inter-group distance was computed:

$$d_{i,j,k} = \frac{\sum_{t=1}^{n[k]} \text{DTW}(\text{drive}_{i,j}, \text{drive}_{k,t})}{n[k]}, \quad D_{i,k} = \frac{\sum_{j=1}^{n[i]} d_{i,j,k}}{n[i]}.$$

We then penalized $D_{i,k}$ by the average group sizes and clustered the resulting inter-group distance matrix with k -means.